INCH-POUND
MIL-PRF-3098/134E
27 July 2005
SUPERSEDING
MIL-PRF-
3098/134D
27 August 1997

PERFORMANCE SPECIFICATION SHEET

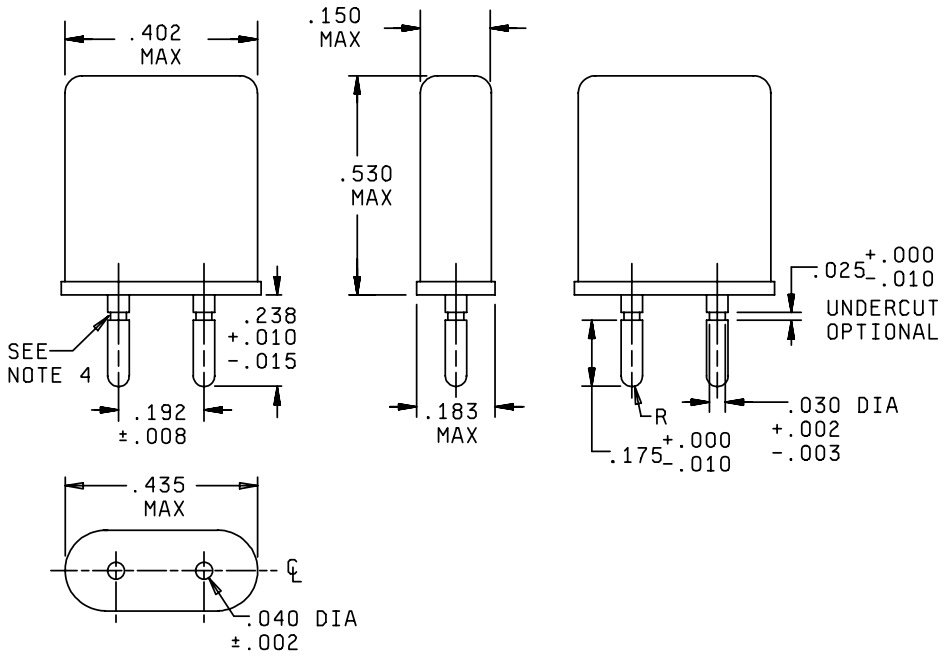
CRYSTAL UNIT, QUARTZ, CR153/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 10.0 MHz to 30.0 MHz; fundamental; noncontrolled; antiresonance.

Inches	mm
.002	.05
.003	.08
.005	.13
.008	.20
.010	.25
.015	.38
.025	.64
.030	.76
.040	1.02
.150	3.81
.175	4.45
.183	4.65
.192	4.88
.238	6.05
.402	10.21
.435	11.05
.530	13.46



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.
4. The pin undercut may be omitted.

FIGURE 1. Crystal unit - CR153/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 10.0 MHz to 30.0 MHz, inclusive.

Frequency tolerance :

Operating temperature: ±10 ppm.

Equivalent resistance: 10 ohms, maximum.

Mode of oscillation: Fundamental.

Temperature range:

Operable: 0°C to +60°C, inclusive.

Operating: -55°C to +105°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Antiresonance load capacitance: 20.0 pF ±0.1 pF.

C_1=24 pF ±10 percent.

C_0/C_1 ratio: 245 ±20 percent.

Frequency stability $\frac{\Delta F}{F}$: ≤3 ppm over the temperature range 0°C to +60°C, when measured at series resonance.

Shock (specified pulse): 1/

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 201, MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

VERIFICATION:

Qualification inspection: Not applicable for this specification.

Conformance inspection: Group A tests of MIL-PRF-3098 shall be applicable.

1/ Method 213, MIL-STD-202, test condition E, except duration shall be 11 milliseconds. Measurements before and after test. Frequency measured at +40°C ±1°C; and equivalent resonant resistance measured at 27°C ±1°C.

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* Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

* The margins of this specification sheet are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force – 19, 99

Preparing activity:
Army - CR

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DLA - CC

(Project 5955-0811-11)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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